Exercise Solutions: If-Else Statement

1. Check Positive or Negative Number

Solution

```
import java.util.Scanner;

public class PositiveOrNegative {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number: ");
        double number = input.nextDouble();

        if (number > 0) {
            System.out.println("Number is positive");
        } else if (number < 0) {
            System.out.println("Number is negative");
        } else {
            System.out.println("Number is zero");
        }

        input.close();
    }
}
```

Sample Output

```
Input number: 35
Number is positive
```

2. Solve Quadratic Equation

Solution

```
import java.util.Scanner;

public class QuadraticEquation {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input a: ");
        double a = input.nextDouble();
        System.out.print("Input b: ");
        double b = input.nextDouble();
        System.out.print("Input c: ");
        double c = input.nextDouble();

        double discriminant = b * b - 4 * a * c;

        if (a == 0) {
            System.out.println("This is not a quadratic equation.");
        } else if (discriminant > 0) {
            double root1 = (-b + Math.sqrt(discriminant)) / (2 * a);
            double root2 = (-b - Math.sqrt(discriminant)) / (2 * a);
            System.out.println("The roots are " + root1 + " and " + root2);
        } else if (discriminant == 0) {
            double root = -b / (2 * a);
            System.out.println("The root is " + root);
        } else {
            System.out.println("The equation has no real roots.");
        }

        input.close();
    }
}
```

Sample Output

```
Input a: 1
Input b: 5
Input c: 1
The roots are -0.20871215252208009 and -4.7912878474779195
```

3. Find Greatest Among Three Numbers

Solution

```
import java.util.Scanner;

public class GreatestOfThree {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input the 1st number: ");
        int first = input.nextInt();
        System.out.print("Input the 2nd number: ");
        int second = input.nextInt();
        System.out.print("Input the 3rd number: ");
        int third = input.nextInt();

        int greatest;
        if (first >= second && first >= third) {
            greatest = first;
        } else if (second >= first && second >= third) {
            greatest = second;
        } else {
            greatest = third;
        }

        System.out.println("The greatest: " + greatest);
        input.close();
    }
}
```

Sample Output

```
Input the 1st number: 25
Input the 2nd number: 78
Input the 3rd number: 87
The greatest: 87
```

4. Weekday Name from Number

Solution

```
import java.util.Scanner;

public class WeekdayName {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number: ");
        int day = input.nextInt();

        if (day == 1) {
            System.out.println("Monday");
        } else if (day == 2) {
            System.out.println("Tuesday");
        } else if (day == 3) {
            System.out.println("Wednesday");
        } else if (day == 4) {
            System.out.println("Thursday");
        } else if (day == 5) {
            System.out.println("Friday");
        } else if (day == 6) {
            System.out.println("Saturday");
        } else if (day == 7) {
            System.out.println("Sunday");
        } else {
            System.out.println("Invalid input");
        }

        input.close();
    }
}
```

Sample Output

```
Input number: 3
Wednesday
```

5. Compare Floats Up to Three Decimals

Solution

```
import java.util.Scanner;

public class CompareFloats {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input floating-point number: ");
        double first = input.nextDouble();
        System.out.print("Input floating-point another number: ");
        double second = input.nextDouble();

        long firstRounded = Math.round(first * 1000);
        long secondRounded = Math.round(second * 1000);

        if (firstRounded == secondRounded) {
            System.out.println("They are the same");
        } else {
            System.out.println("They are different");
        }

        input.close();
    }
}
```

Sample Output

```
Input floating-point number: 25.586
Input floating-point another number: 25.589
They are different
```